

Deterministic pose-spline dynamics estimator

Estimator structure

pose-only continuous-time spline -> analytic derivatives
pose spline degree: 5 (position and normalized quaternion)
fixed-spline gradient matching -> shared physical parameters and lag
default physical initialization: exact nominal chart origin
previous estimator result loading: explicit opt-in only
shooting nodes: none
continuity constraints: none
sensor/IMU channels in loss: none

Initial estimate

source kind: nominal
source path: None
source mass [kg]: 2.35159759081
selected initial mass [kg]: 2.35159759081

Selected strict-ZOH parameters

mass [kg]: 2.76332418875
inertia [kg m²]:
[0.071227673206, -0.00205598936875, 0.000151379334808]
[-0.00205598936875, 0.152541470205, 0.000640747269435]
[0.000151379334808, 0.000640747269435, 0.222793681867]
principal moments [kg m²]: [0.0711755363771, 0.152587647386, 0.222799641515]
inertia triangle margin [kg m²]: 0.000963542247835
CoG offset [m]: [-0.00663161744515, 0.000646509369453, -0.0169246331634]
rotor force effectiveness: [1.01394245002, 1.01764363594, 0.967518414904, 1.00168629891]
lag [s]: 0.041514156863
joint dynamics loss: 0.431909397151
soft-prior cost: 0.0503352330189

Smooth physical coordinates

log_mass_scale	0.161339448158
log_second_moment_cholesky_xx_scale	0.428998757363
log_second_moment_cholesky_yy_scale	0.0458229016676
log_second_moment_cholesky_zz_scale	0.00157268707625
normalized_second_moment_cholesky_yx_offset	0.020750069629
normalized_second_moment_cholesky_zx_offset	-0.0042002163307
normalized_second_moment_cholesky_zy_offset	-0.0321889901428
cog_offset_x_m	-0.00663161744515
cog_offset_y_m	0.000646509369453
cog_offset_z_m	-0.0169246331634
force_effectiveness_contrast_1	-0.00257644662559
force_effectiveness_contrast_2	0.0397542320317
force_effectiveness_contrast_3	-0.00194553035075

Spline and collocation

spline degree: 5
collocation step [s]: 0.01

Bag failure_1

selected spline knot spacing [s]: 0.2
dynamics loss: 0.62719210085
estimated forward position RMSE [m]: 43.1028679899

estimated + external wrench position RMSE [m]: 0.806311619264
external wrench improves estimated free rollout: True
nominal forward position RMSE [m]: 68.7734508548
estimated improves nominal: True
residual wrench mean: [0.188128 0.039747 0.029011 -0.028326 -0.011455 -0.022919]
residual wrench RMSE: [0.971552 0.957233 1.426551 0.159144 0.664478 0.260582]
residual wrench trend/s: [1.250227e-01 -1.109141e-02 -5.558662e-01 -1.119819e-02
-5.770317e-02
3.042122e-04]
residual wrench cumulative impulse: [1.063688 0.246719 0.159423 -0.165394 -0.03806
-0.127146]

Bag failure_2

selected spline knot spacing [s]: 0.2
dynamics loss: 0.236626693451
estimated forward position RMSE [m]: 50.2396039059
estimated + external wrench position RMSE [m]: 0.778601976086
external wrench improves estimated free rollout: True
nominal forward position RMSE [m]: 66.3356799895
estimated improves nominal: True
residual wrench mean: [0.053185 -0.304956 -0.001855 0.028605 0.034661 0.046538]
residual wrench RMSE: [0.544606 0.472273 0.847358 0.070878 0.30442 0.39373]
residual wrench trend/s: [-0.067324 0.030329 -0.317856 -0.007142 0.054269 0.002791]
residual wrench cumulative impulse: [0.327201 -1.691241 -0.050226 0.161195 0.199807
0.213931]

Smoothstep continuation

width=0.5: delay=0.0391275156s, cost=0.482173850468, nfev=17
width=0.2: delay=0.0387613646s, cost=0.482218385371, nfev=6
width=0.05: delay=0.0405141569s, cost=0.482292000084, nfev=21

Strict-ZOH polish

delay=0.0365141569s: screening=0.486867018956, refined=nan, selected=False
delay=0.0375141569s: screening=0.482965816031, refined=nan, selected=False
delay=0.0385141569s: screening=0.482446795282, refined=nan, selected=False
delay=0.0395141569s: screening=0.482314125291, refined=0.482314122205, selected=False
delay=0.0405141569s: screening=0.482292581908, refined=0.482292581672, selected=False
delay=0.0415141569s: screening=0.482244639268, refined=0.48224463017, selected=True
delay=0.0425141569s: screening=0.484083793234, refined=nan, selected=False
delay=0.0435141569s: screening=0.483653862672, refined=nan, selected=False
delay=0.0445141569s: screening=0.483911523991, refined=nan, selected=False